AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102

Final Examination
Fall Semester 2025/2026

Date: 15/Jan/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2hrs

Instructions:

- Total Marks: **100**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I- Answer the following (MCQ). [50 Marks

1 Magnitude of displacement from initial position to final position is the
A. straight line B. curved line C. circle D. total distance

2 Average acceleration of a body during time interval 't' is given by slope of its
A. velocity-speed graph B. velocity-time graph
C. acceleration-time graph D. velocity-displacement graph

3 Potential difference between ends of conductor maintained by battery is
A. different B. constant C. zero D. one

4 A man travels to a planet that has twice the radius of the earth and twice the mass. His weight, compared to his weight on earth, is
A. Double B. Triple C. Half D. No Enough data

5 Which of these statements comparing electric and gravitational forces is correct?

A The electric and gravitational forces exerted by two particles on each other are inversely proportional to the separation of the particles.

B None of these

C The electric force exerted by one planet on another is typically stronger than the gravitational force exerted by that same planet on the other.

D The direction of the electric force exerted by one point particle on another is always the same as the direction of the gravitational force exerted by that particle on the other.

6 The intensity of sunlight that reaches Earth's atmosphere is 1000 W/m^2. If Jupiter is 6times as far from the Sun as Earth, the intensity of the sunlight that reaches Jupiter is

A 27.7W/m2 B 269.2W/m2 C 51.7W/m2 D 1.52 W/m2

7 The change of momentum of an object is equal to

A Force*time C Force/time
B Impulse/time D Mas*velocity

8 The rate of change of momentum of an object is proportional to

A Weight of the body C Velocity of the body
B Net force applied on the body D non of these

9 The intensity of an isotropic sound wave is

A directly proportional to the square of the distance from the source
B inversely proportional to the square of the distance from the source
C directly proportional to the distance from the source
D inversely proportional to the distance from the source

10 How does the resistance of a piece of conducting wire change if both its length and diameter are doubled?

A 4 times original B 1/2 times original C 2 times original D Remains the same

11 The electrical properties of copper and rubber are different because

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A Many electrons are free to move in copper but nearly all are bound to molecules in rubber.

B The positive charges are free to move in rubber but are stationary in copper.

C The positive charges are free to move in copper and stationary in rubber.

D Many electrons are free to move in rubber but nearly all are bound to molecules in copper.

12 The junction rule tells us that the same current flows out of any battery with emf equal to

A Current flows out of other battery connected in parallel

B Total equivalent resistor Req of circuit/ emf of battery

C Current flows in the same battery

D emf of battery * total equivalent resistor Req of circuit

13 Two objects of masses 200g and, 300g are moving along the same line and direction with velocities of 15m/s and 10m/s respectively. They collide and after the collision the first object moves at a velocity of 7m/s. the velocity of the second object is:

A 14.33 m/s

B 20m/s

C 15.33 m/s

D 17m/s

14 When rubbing two insulators together, the one that gets a positive charge has had.

A Electrons added.

B Electrons removed

C Protons removed.

D Protons added.

15 Two charged particles attract each other with a force of magnitude F acting on each. If the charge of one is doubled and the distance separating the particles is also doubled, the force acting on each of the two particles has magnitude

A F/2

B 2F

C F/4

D F

16. The force required to accelerate an object of mass 4 kg at a rate of 3 m/s^2 is:
a) 12 N b) 2 N c) 3 N d) 5 N

17. The oscillations of particles in a longitudinal wave occur:
a) Perpendicular to the direction of wave propagation
b) In the same direction as the wave motion
c) At right angles to the wave amplitude
d) In random directions

18. A wave passes through a medium at a frequency of 20 Hz. What is the period of the wave?
a) 20 seconds b) 0.05 seconds c) 0.2 seconds d) 2 seconds

19. When two positively charged objects are brought close together, they will:
a) Attract each other b) Repel each other

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c) Cancel each other's charge d) Have no effect on each other

20. Coulomb's law describes the relationship between electric force, charges, and:
a) Time b) Distance c) Speed d) Energy

21. If the distance between two point charges is doubled, the force between them will:
a) Quadruple b) Half c) Double d) Remain unchanged

22. The opposition offered by a material to the flow of electric current is called:
a) Conductivity b) Voltage c) Resistance d) Current

23. Kirchhoff's first law is related to:
a) Conservation of charge b) Conservation of energy
c) Conservation of momentum d) Conservation of voltage

24. What property of an object determines whether it gains or loses electrons?
a) Mass b) Volume c) Electric charge d) Electric field strength

25. The resistance of a wire depends on its:
a) Length only b) Cross-sectional area only
c) Material and length d) Material and cross-sectional area

Part II: Answer the following questions. [50Marks]

Question 1: (25 Marks)

a) A wave travels at 175 m/s along the x-axis. If the period of the periodic vibrations of the wave is 3.00 milliseconds, then what is the wavelength of the wave? (10 marks)

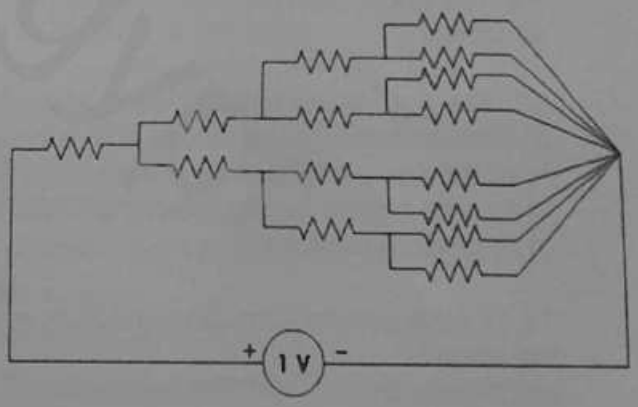
b) Two carbon cylinders have radii r and 2r, and they both have length L. What is the resistance of the first if the resistance of the second is 12Ω? (15 marks)

Question2: (25 Marks)

a) Define the ohm. (5 Marks)

b) All the resistances in figure shown below are 1 Ω each. Calculate the total resistance (10 Marks)

c) Calculate the value of electric current in Ampere through the battery. (10 Marks)



End of questions

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